Prediction of PVT Behaviour and Thermodynamic Properties (H, S) of Real Gases using Cubic EOS

(van der Waals, Redlich/Kwong, Soave/Redlich/Kwong and Peng/Robinson)

CHL2002: Chemical Engineering Thermodynamics, Prof. S. K. Gupta

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Github: `https://github.com/arushijain1182/CHL2002_Cubic_EOS_Project`

1. Introduction and importance

Real gases deviate from $PV = RT$, and cubic equations of state (EOS) are the standard way to correct for this. They are **analytically tough to use**: the EOS is cubic in V, so finding V from (T, P) means solving a cubic and choosing the right root (vapour, liquid or unstable); finding T from (P, V) needs iteration; the saturation pressure needs an iterative equal-fugacity calculation; and H, S need derivatives of $a(T)$. By hand this is slow, and chart reading gives only approximate values. **The program does all of this in one run: given a substance, an EOS and any two of T, P, V, it returns the third, together with Z, density, phase, fugacity coefficient, H, S, and the deviation from the ideal gas.**

2. Theory

2.1 General cubic EOS

All four equations are special cases of one form, written two equivalent ways.

As discussed in class (u, w form):

$$P = \frac{RT}{V-b} - \frac{a(T)}{V^2 + u\,bV + w\,b^2} \tag{1}$$

As in the textbook (Smith, Van Ness, Abbott & Swihart, Eq. 3.41; σ, ϵ form):

$$P = \frac{RT}{V-b} - \frac{a(T)}{(V+\epsilon b)(V+\sigma b)} \tag{2}$$

Expanding $(V+\epsilon b)(V+\sigma b) = V^2 + (\sigma+\epsilon)bV + \sigma\epsilon\, b^2$ gives the relation between them:

$$u = \sigma + \epsilon, \qquad w = \sigma\epsilon \tag{3}$$

$$b = \Omega\frac{RT_c}{P_c}, \qquad a(T) = \Psi\,\alpha(T_r,\omega)\frac{R^2T_c^2}{P_c} \tag{4}$$

Here b accounts for molecular size and $a(T)$ for attraction. Ω, Ψ, u, w (σ, ϵ) are fixed numbers of each EOS (Table 1); the fluid enters only through T_c, P_c (in b, a, $T_r = T/T_c$, $P_r = P/P_c$) and the acentric factor ω (in α). These were **taken from Table B.1 of the textbook**. Do not confuse ω with w.

EOS	u	w	σ	ϵ	Ω	Ψ	$\alpha(T_r,\omega)$
vdW	0	0	0	0	1/8	27/64	1 (a constant)
RK	1	0	1	0	0.08664	0.42748	$T_r^{-1/2}$
SRK	1	0	1	0	0.08664	0.42748	$[1+(0.480+1.574\omega-0.176\omega^2)(1-\sqrt{T_r})]^2$
PR	2	-1	$1+\sqrt{2}$	$1-\sqrt{2}$	0.07780	0.45724	$[1+(0.37464+1.54226\omega-0.26992\omega^2)(1-\sqrt{T_r})]^2$

Table 1: Constants of the four EOS (Table 3.1 of the text; u, w as used in class).

2.2 Volume, phase and saturation

With $\beta = \Omega P_r/T_r$, $q = \Psi\alpha/(\Omega T_r)$ and $Z = PV/RT$, Eq. (1) becomes a cubic in Z (real roots with $Z > \beta$ are kept):

$$Z^3 + [u\beta - (1+\beta)]Z^2 + [w\beta^2 - (1+\beta)u\beta + q\beta]Z - [(1+\beta)w\beta^2 + q\beta^2] = 0 \tag{5}$$

Above T_c there is one root. Below T_c there can be three: smallest = liquid, largest = vapour, middle = unstable. The phase is chosen by comparing P with P^{sat}, found from equal fugacity of the two phases:

$$\ln\varphi = Z - 1 - \ln(Z-\beta) - qI, \qquad I = \frac{1}{\sigma-\epsilon}\ln\frac{Z+\sigma\beta}{Z+\epsilon\beta} \quad \text{(vdW: } I = \beta/Z) \tag{6}$$

$P(T, V)$ is direct from Eq. (1); $T(P, V)$ is found by bisection, since P rises monotonically with T at fixed V.

2.3 Enthalpy and entropy

Residual properties (text Eqs. 13.75–13.76):

$$\frac{H^R}{RT} = Z - 1 + \left[\frac{d\ln\alpha}{d\ln T_r} - 1\right] qI, \qquad \frac{S^R}{R} = \ln(Z-\beta) + \frac{d\ln\alpha}{d\ln T_r}\, qI \tag{7}$$

with $d\ln\alpha/d\ln T_r = 0$ (vdW), $-1/2$ (RK) and $-\kappa\sqrt{T_r}/[1+\kappa(1-\sqrt{T_r})]$ (SRK, PR; κ is the polynomial in ω of Table 1). The ideal-gas part uses $C_P^{ig}/R = A + BT + CT^2 + D/T^2$ (**Table C.1** of the text):

$$H = \int_{T_0}^{T} C_P^{ig}\, dT + H^R, \qquad S = \int_{T_0}^{T} \frac{C_P^{ig}}{T}\, dT - R\ln\frac{P}{P_0} + S^R \tag{8}$$

The reference ($H = S = 0$) is an ideal gas at $T_0 = 298.15$ K, $P_0 = 1$ bar (changeable); only differences between states are physically meaningful. Latent heat: $\Delta H^{vap} = H^R_{vapour} - H^R_{liquid}$ at P^{sat}.

3. Data used

Quantity	Source	Used for
T_c, P_c, ω, molar mass (89 species)	**Table B.1** of the textbook	b, $a(T)$, T_r, P_r, α
Ideal-gas C_P constants A–D (41 species)	Table C.1 of the textbook	H, S (ideal-gas part)
EOS constants $\Omega, \Psi, \sigma, \epsilon, \alpha$	Table 3.1 of the textbook	Eqs. (1)–(4)

Table 2: All substance data come from the textbook tables and are stored in `cubic_eos.py`.

The substance is chosen by name (e.g. "methane", "n-butane", "carbon dioxide"; case and spaces/hyphens ignored). A fluid not in Table B.1 can be entered by giving T_c, P_c, ω. Table C.1 lists C_P^{ig} for only 41 of the 89 species. For the other 48 no approximation is made: the program still gives P, V, T, Z, phase, φ and the residuals H^R, S^R, but not total H and S, unless the user supplies the constants A–D (`--cp A B C D`; the website has an input box for this).

4. How to use the program

File	Purpose
`cubic_eos.py`	Program: data, four EOS, solver, H and S, plots, command line
`app.py`	Website (Streamlit): substance → EOS → two of T, P, V → results, graph, deviation from ideal gas
`test_cubic_eos.py`	21 automated checks against the textbook
`requirements.txt`, `README.md`	Install list, instructions

Table 3: Repository contents.

Website: `pip install -r requirements.txt`, then `streamlit run app.py`. Choose the substance, then the EOS, then the known pair (T, P), (T, V) or (P, V) with units. The page shows P, T, V, Z, phase, H, S, a P–V and Z–P graph against the ideal gas, and a closing line with the percentage deviation from the ideal gas.

Command line:

- `python cubic_eos.py props n-butane --T 350 --P 5 --eos PR` (V, Z, phase, H, S)
- `python cubic_eos.py p methane --T 323.15 --V 120 --eos RK` (P from T, V)
- `python cubic_eos.py t ammonia --P 23.82 --V 1021.2 --eos PR` (T from P, V)
- `python cubic_eos.py sat n-butane --T 350` (saturation); also `compare`, `plot`, `list-species`

5. Validation and results

5.1 Worked examples from the text

Example 3.9 (n-butane, 350 K, 9.4573 bar): the program reproduces the book's values for all four EOS.

Molar volume, cm^3/mol	vdW	RK	SRK	PR	Exp.
Vapour (book = program)	2667	2555	2520	2486	2482
Liquid (book = program)	191.0	133.3	127.8	112.6	115.0

Table 4: Example 3.9; program output equals the book to the printed digits.

Case (property)	Exp.	Ideal gas	vdW	RK	SRK	PR
Ex. 3.10: n-butane 510 K, 25 bar (V)	1480.7	1696.1 (+14.5%)	1463.2 (−1.2%)	1462.2 (−1.3%)	1486.3 (+0.4%)	1454.7 (−1.8%)
Ex. 3.11: methane 323.15 K, 120 cm^3/mol (P)	196.5	223.9 (+13.9%)	189.3 (−3.7%)	198.4 (+1.0%)	204.4 (+4.0%)	191.9 (−2.3%)
Ex. 3.12: ammonia 338.15 K, 1021.2 cm^3/mol (P)	23.82	27.53 (+15.6%)	24.50 (+2.9%)	23.83 (0.0%)	23.58 (−1.0%)	23.37 (−1.9%)

Table 5: Comparison with experimental values quoted in the text (units: cm^3/mol for V, bar for P).

The ideal gas is 14–16% off; the cubic equations are mostly within a few percent, and no single equation is best everywhere.

5.2 Checks on H and S

The text has no worked cubic-EOS enthalpy example, so H^R and S^R were checked independently (automated in `test_cubic_eos.py`):

- **Gibbs–Helmholtz:** $H^R/RT = -T(\partial \ln \varphi/\partial T)_P$, agreement better than 0.002% for all four EOS.
- **Clapeyron:** ΔH^{vap} from residual properties equals $T\Delta V\, dP^{sat}/dT$ (ammonia, 310 K; RK, SRK, PR) within 0.02%.
- **Lee/Kesler tables** (D.5–D.12, $\omega = 0.012$, 64 gas states): SRK within $0.05\,RT_c$ in H^R and $0.02\,R$ in S^R; PR within $0.10\,RT_c$ and $0.03\,R$.
- **Limits:** $H^R, S^R \to 0$ as $P \to 0$.

5.3 Example output and saturation properties

CO_2, 350 K, 80 bar	Z	V, cm^3/mol	H^R, J/mol	S^R, J/(mol K)	H, J/mol	S, J/(mol K)
vdW	0.7137	259.6	−2241	−4.30	−221	−34.50
RK	0.7212	262.4	−2682	−5.50	−661	−35.69
SRK	0.7502	272.9	−2936	−6.40	−916	−36.60
PR	0.7217	262.5	−3041	−6.43	−1020	−36.63

Table 6: Program output. Ideal gas: $Z = 1$, $V = 363.8$ cm^3/mol, $H^R = S^R = 0$ (about 39% volume error).

n-butane, 350 K	P^{sat}, bar	Error vs exp.	V_{liq}, cm^3/mol	ΔH^{vap}, kJ/mol	ΔS^{vap}, J/(mol K)
vdW	16.60	+75.5%	186.1	8.17	23.4
RK	11.41	+20.7%	132.8	14.69	42.0
SRK	9.59	+1.4%	127.8	17.55	50.1
PR	9.47	+0.1%	112.6	17.47	49.9

Table 7: Saturation properties (experimental $P^{sat} = 9.4573$ bar, $V_{liq} = 115.0$ cm^3/mol).

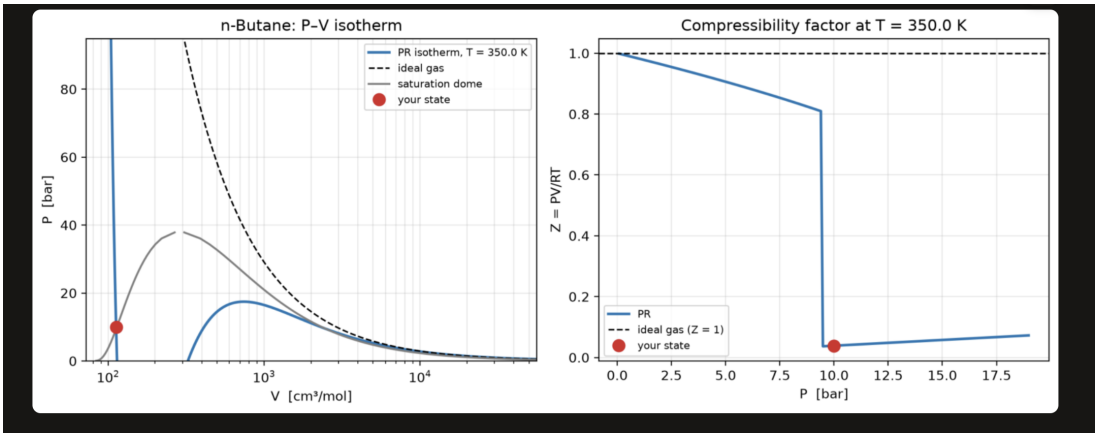


Figure 1: Graphs produced by the website for n-butane at 350 K and 10 bar with PR. Left: P–V isotherm against the ideal gas, with the saturation dome and the calculated state. Right: Z against P; the state is liquid, just above $P^{sat} = 9.47$ bar, where Z drops sharply.

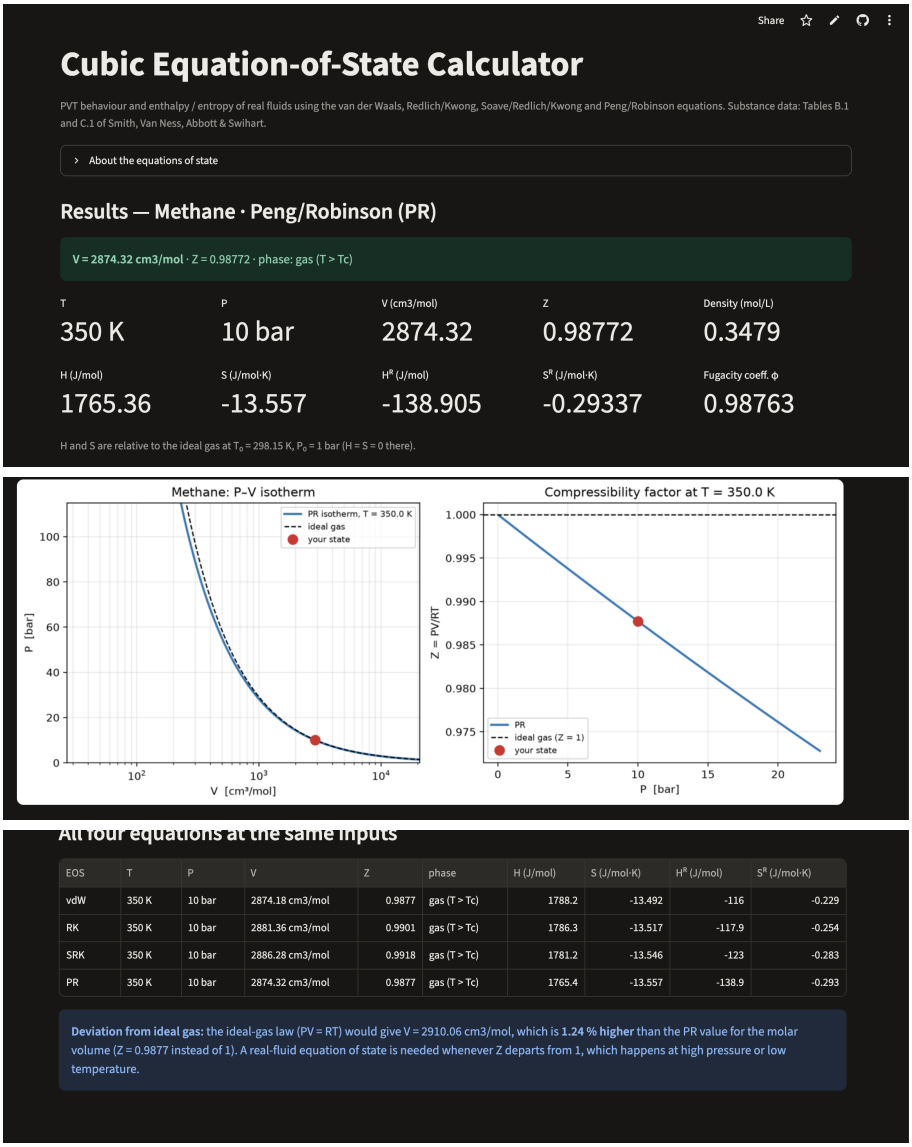


Figure 2: Website output for methane at 350 K and 10 bar with PR. Top: substance, EOS and input choices, with V, Z, density, H, S, H^R, S^R and fugacity coefficient. Middle: P–V isotherm and Z–P plot against the ideal gas. Bottom: the same state from all four EOS, and the deviation from the ideal gas (ideal gas gives V 1.24% higher than PR).

6. Discussion and limitations

- **Vapour:** all four EOS work well at low-to-moderate reduced pressure; vdW is the least reliable overall (P^{sat} of n-butane 76% too high).
- **Liquids:** vdW, RK and SRK predict saturated-liquid volumes poorly (n-butane, 350 K: vdW 191 vs 115 cm^3/mol experimental); PR is better.
- **Scope:** pure fluids only. H, S need ideal-gas C_P data (Table C.1 or user-supplied A–D); otherwise only H^R, S^R are given. Quantum gases (H_2, He) need the effective critical constants of Eqs. 3.54–3.56.

7. Conclusion

The program replaces hand iteration and chart reading for the four cubic equations taught in class, reproduces the textbook examples, satisfies the thermodynamic consistency relations for H and S, and quantifies the departure from ideal-gas behaviour (14–39% in the cases above).

8. References

J. M. Smith, H. C. Van Ness, M. M. Abbott, M. T. Swihart, *Introduction to Chemical Engineering Thermodynamics*, 8th ed., McGraw-Hill, 2018 (Ch. 3, 6, 13; Appendices B, C).